SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1502

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 4

Code of Conduct

I **G. Umesh Chandra Yadav/192525424**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.Umesh

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

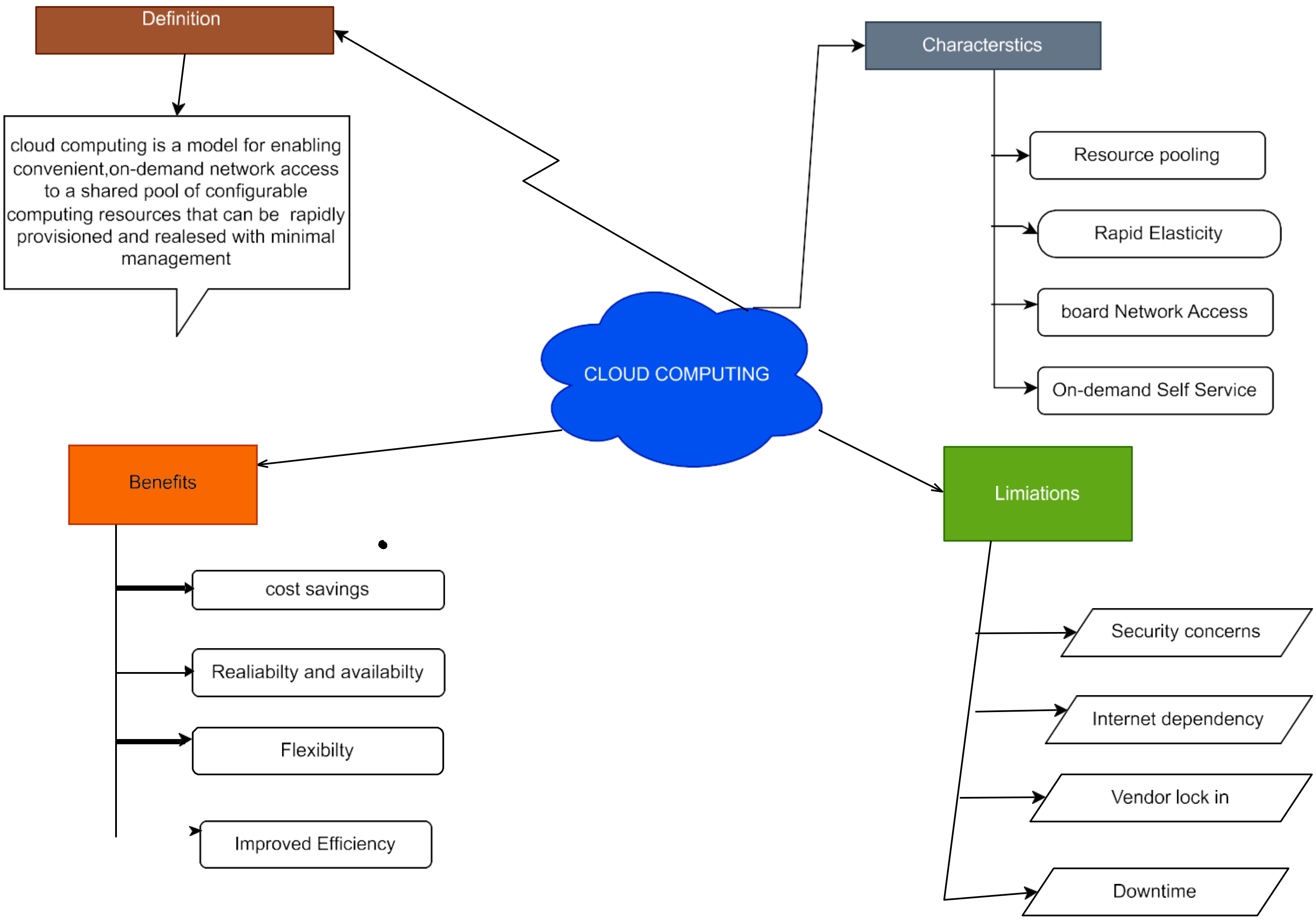
Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect

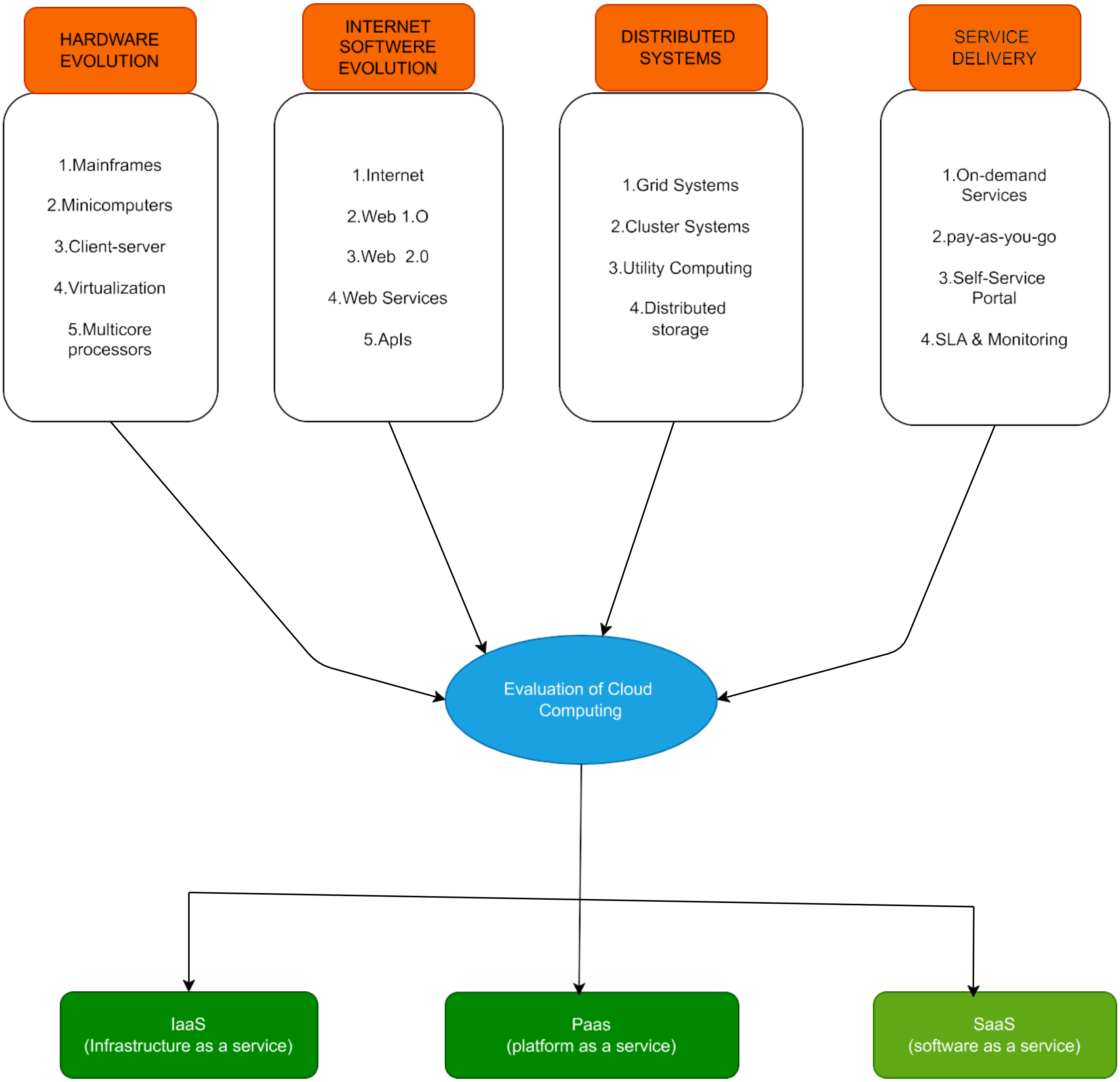
		accurate and meaningful			
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

1. Evaluate the effectiveness of your concept map in representing cloud computing principles. Which Design decisions most strongly support your solution, and why?
2. Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?
3. Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?
4. Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?
5. Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

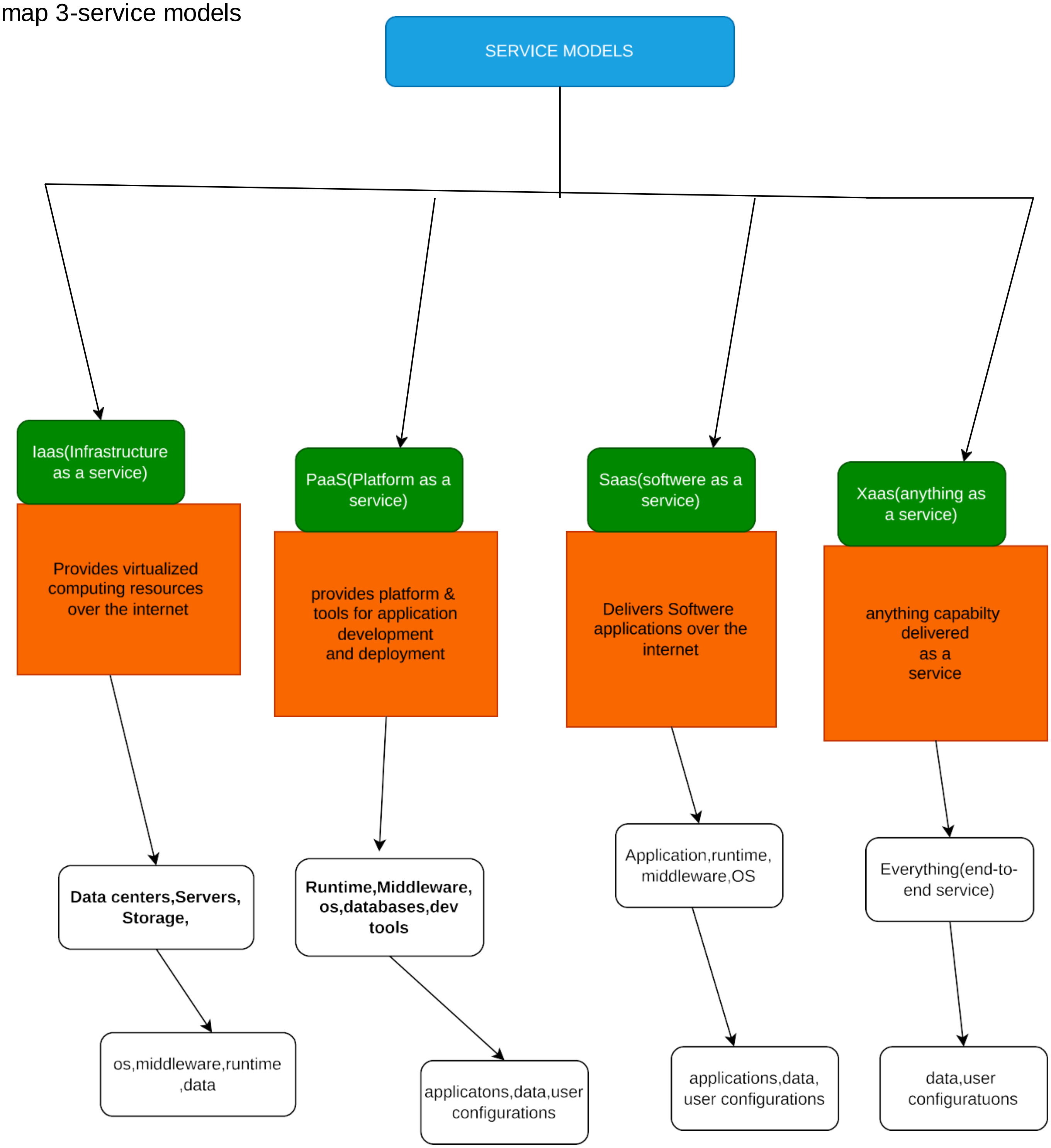
concept map 1-cloud computing fundamentlas



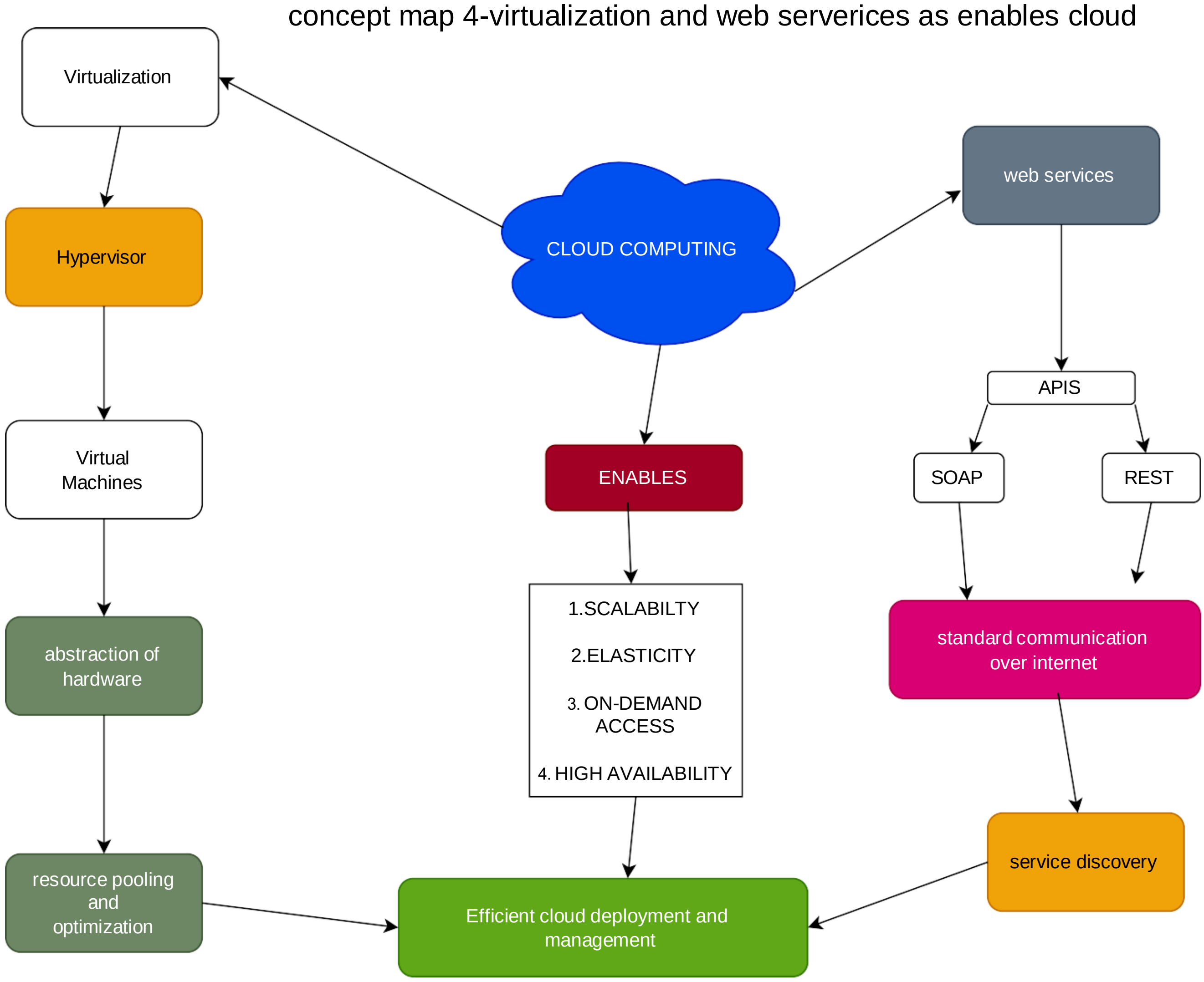
concept map 2- evalution of cloud computing



concept map 3-service models



concept map 4-virtualization and web serverices as enables cloud



1. **Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?**

   The concept map clearly shows the main cloud computing components and their relationships. It includes users, cloud services, storage, APIs, security, and resource management. The layer design makes the system easy to understand. Connecting all components with arrows shows the data flow. This design improves clarity, organization, and understanding of cloud architecture.

2. Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

Scalability allows the system to handle more users by adding resources. Elasticity automatically increases or decreases resources based on demand. Resource management efficiently allocates CPU, memory, and storage. Without these features, the system may become slow, overloaded, and expensive. Users may experience delays, downtime, and poor performance.

3.Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?

APIs enable communication between applications and cloud services. Webservices allow data exchange over the internet. Cloud services provide computing, storage, and database facilities. These services improve flexibility, automation, and integration. They reduce development time and increase system efficiency.

4.Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

Users access applications through APIs and web services. Applications communicate with cloud servers and databases. Security protects data during communication. Resource management balances workloads for better performance. These interactions improve reliability, faster response, and a better user experience.

5. Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Creating the concept map improved my understanding of cloud computing architecture. I learned how different cloud components work together. The map helped visualize data flow and service interactions. If redesigned, I would add more security features, monitoring tools, and detailed service models (IaaS, PaaS, SaaS). This would make the concept map more complete and easier to understand